12/9/2025 2.21(a)

$$x[n] = \alpha^n u[n] \quad \left.\begin{matrix} \\ \end{matrix}\right\} \quad \alpha \neq \beta \qquad 0 < \alpha, \beta < 1$$
$$h[n] = \beta^n u[n] \quad \left.\begin{matrix} \\ \end{matrix}\right\}$$

$$y[n] = x[n] * h[n]$$
$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

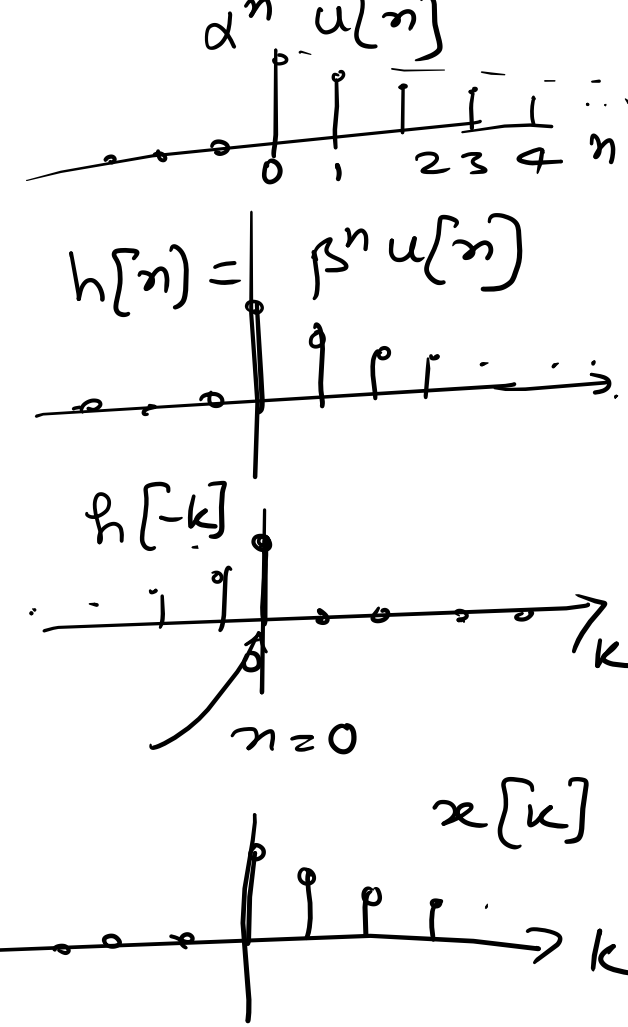
$n < 0$, $y[n] = 0$

$\boxed{n = -1}$

$n \geq 0$

$$y[n] = \sum_{k=0}^{n} \alpha^k (\beta)^{n-k}$$
$$= \beta^n \sum_{k=0}^{n} \left(\frac{\alpha}{\beta}\right)^k$$

$$\boxed{y[n] = \frac{\beta^n \left(1 - (\alpha/\beta)^{n+1}\right)}{1 - (\alpha/\beta)}}$$



$$y[n] = \begin{cases} 0 & n < 0 \\ \dfrac{\beta^{n+1} - \alpha^{n+1}}{\beta - \alpha} & n \geq 0 \end{cases}$$

or $$y[n] = \left(\frac{\beta^{n+1} - \alpha^{n+1}}{\beta - \alpha} \right) u[n]$$

(b) $$h[n] = \alpha^n u[n] = x[n]$$

$n < 0 \Rightarrow y[n] = 0$

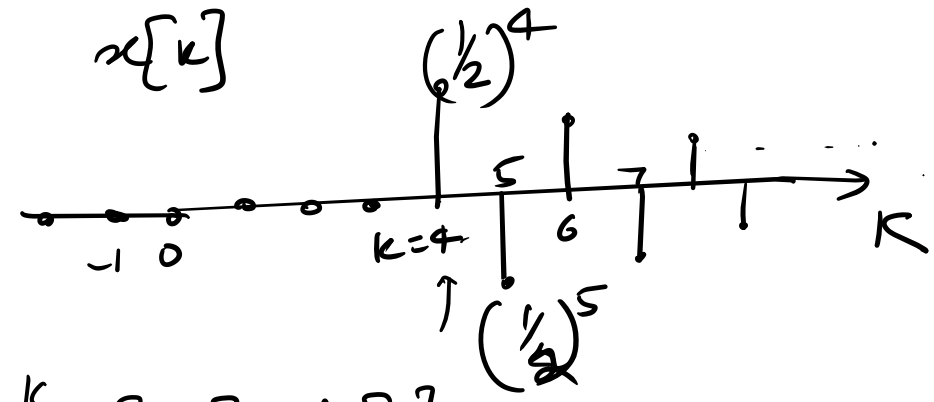
For $n \geq 0$; $y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \sum_{k=0}^{n} \alpha^k \alpha^{n-k}$

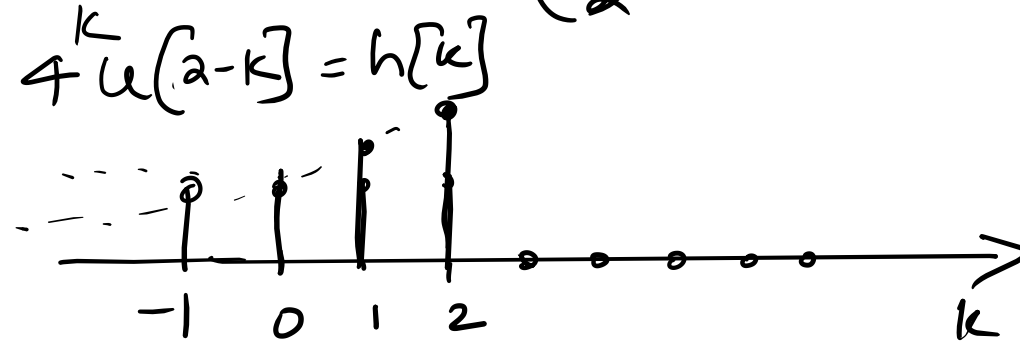
$$= \alpha^n \sum_{k=0}^{n} 1 = (n+1)\alpha^n$$

$\therefore$ $y[n] = (n+1)\alpha^n u[n]$

(c) $x[n] = \left(-\frac{1}{2}\right)^n u[n-4]$

$h[n] = 4^n u[2-n]$





Until $n-2 \leq 4$

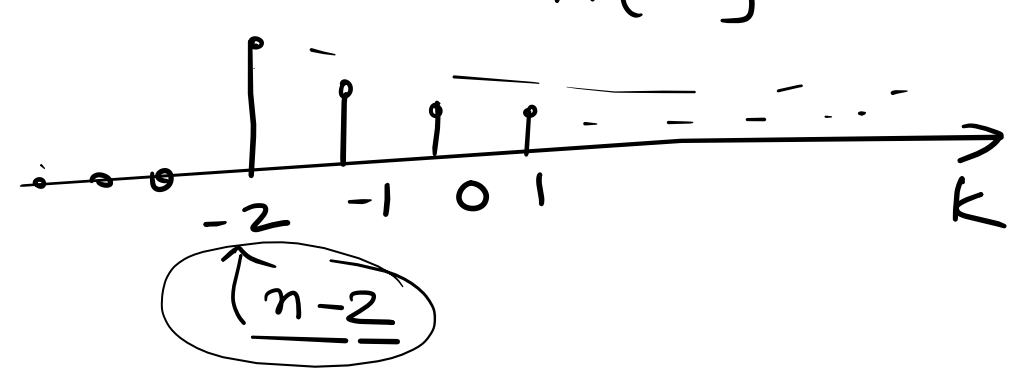
$$y[n] = \sum_{k=4}^{\infty} x[k]\, h[n-k]$$

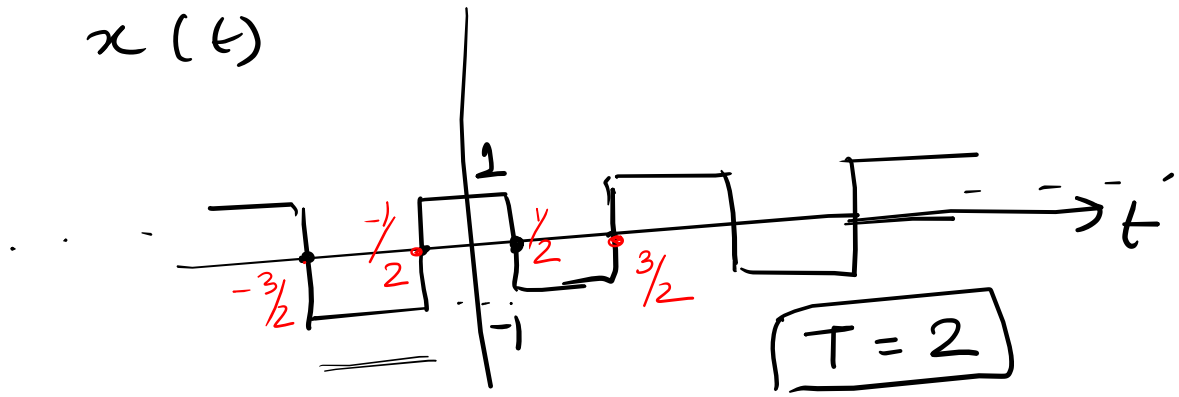
$$= \sum_{k=4}^{\infty} \left(-\frac{1}{2}\right)^k 4^{n-k}$$

$$= 4^n \sum_{k=4}^{\infty} \left(-\frac{1}{8}\right)^k$$

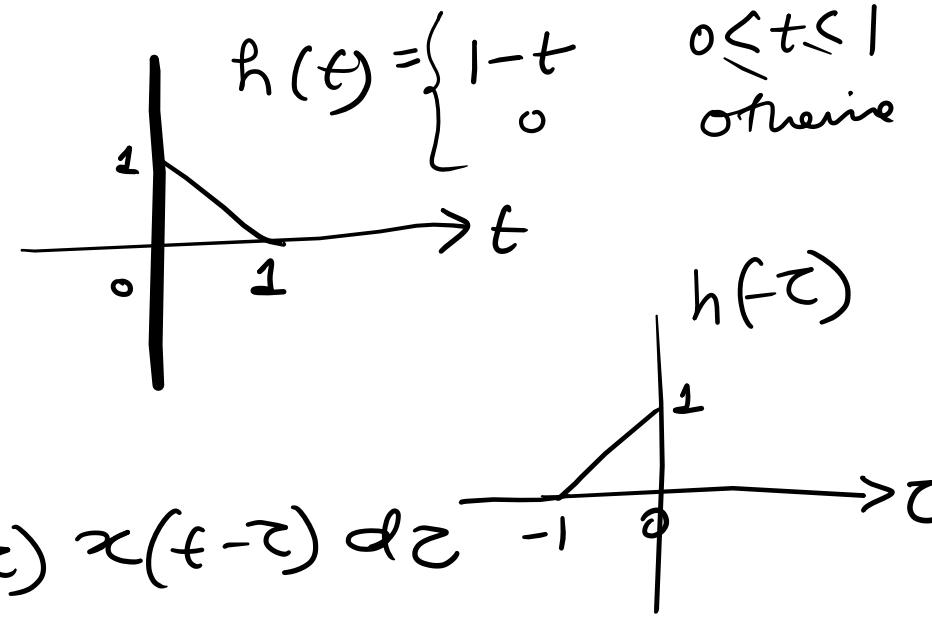
$n-2 > 4$

$$y[n] = 4^n \sum_{k=n-2}^{\infty} \left(-\frac{1}{8}\right)^k$$





periodic

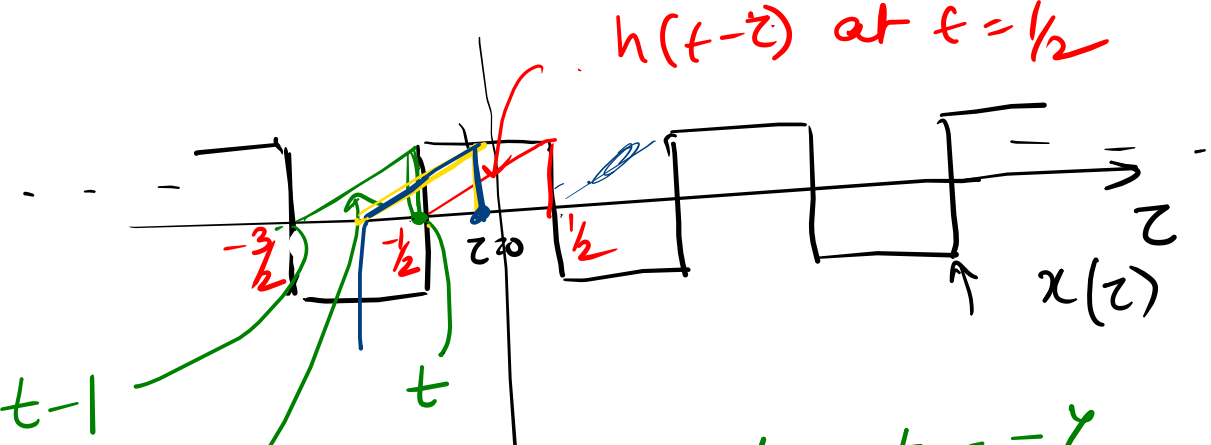


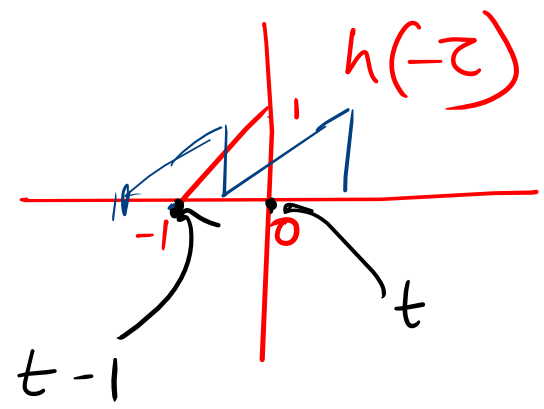
$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$y(t+T) = \int_{-\infty}^{\infty} h(\tau)\, x(t+T-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau = y(t)$$

Given an LTI system a periodic i/p leads to a periodic o/p.
with the same time period.





between $-1/2$ to $3/2$

Let us compute $y(t)$

$$y(t) = \int_{t-1}^{-1/2} h(t-\tau)\,d\tau + \int_{-1/2}^{t} h(t-\tau)\,d\tau$$

$$y(t) = \int_{t-1}^{1/2} \underbrace{h(t-\tau)}_{=1-(t-\tau)}\,d\tau + \int_{1/2}^{t} -h(t-\tau)\,d\tau$$

$$-\frac{1}{2} \le t \le \frac{1}{2}$$

$$\frac{1}{2} \le t \le \frac{3}{2}$$

Computing $y(t)$ in one time period. After that it repeats.

C.T. System — Linear constant coefficient differential eqn.

Def, there is an causal LTI system & the condition of initial rest holds true.

that means if $x(t) = 0$ for $t < t_0$

then $y(t) = 0$ for $t < t_0$

> For N=1
> if $x(t)=0, t<t_0$
> $y(t)=0, t<t_0$
> $\frac{d}{dt}y(t)=0, t<t_0$

Such causal LTI system can be represented in general as —

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

$N \equiv$ order of the system

Let $N = 1$ & $M = 0$

$$a_1 \frac{d}{dt} y(t) + a_0 y(t) = b_0 x(t)$$

set of initial conditions

$$\Rightarrow \frac{d}{dt} y(t) + (a_0/a_1) y(t) = (b_0/a_1) x(t)$$

D.T. System

Causal LTI systems can be represented by linear constant coefficient difference eqn.

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k]$$

Let $N = 1$ & $M = 0$

$$a_1\, y[n-1] + a_0\, y[n] = b_0\, x[n]$$

$$y[n-1] + \left(a_0/a_1\right) y[n] = \left(b_0/a_0\right) x[n]$$

Ex Let us given a system — $\int$ N=1 & M=0

$$y[n] + \left(\frac{1}{2}\right) y[n-1] = x[n]$$

Compute the impulse response of the system & plot it

if $\boxed{x[n] = \delta[n]}$, then $y[n] = h[n]$
= impulse response of the system.

$$y[n] = x[n] - \left(\frac{1}{2}\right) y[n-1]$$

$$\boxed{y[-1] = 0 - \left(\frac{1}{2}\right) y[-2]}$$

$$y[0] = 1 - 0 = 1 \quad \because x[n] = \delta[n]$$

$$y[1] = 0 - \left(\frac{1}{2}\right) y[0] = \left(-\frac{1}{2}\right)$$

$$y[2] = 0 - \frac{1}{2} \times y[1] = \left(-\frac{1}{2}\right)^2$$

$$y[n] = \left(-\frac{1}{2}\right)^n u[n]$$

$$= h[n]$$

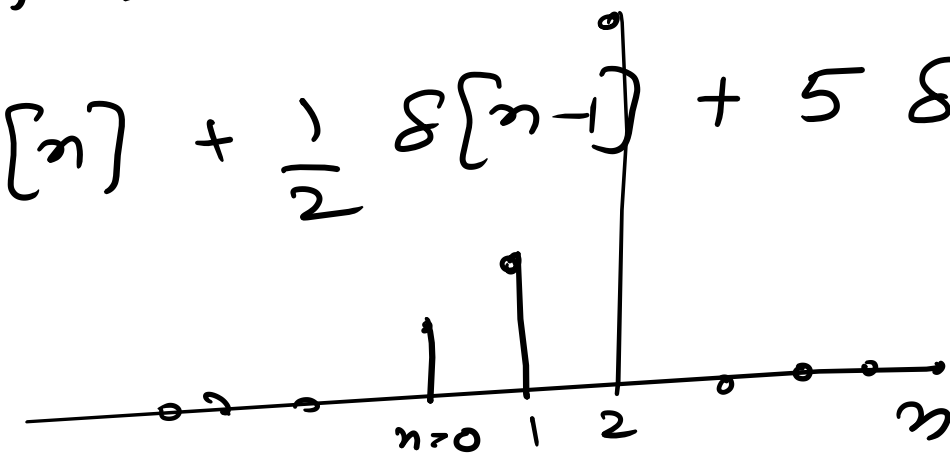
IIR system
Infinite Impulse response system

Ex $y[n] = \frac{1}{3}x[n] + \frac{1}{2}x[n-1] + 5x[n-2]$

$h[n] = ?$

if $x[n] = \delta[n]$, then $y[n] = h[n]$

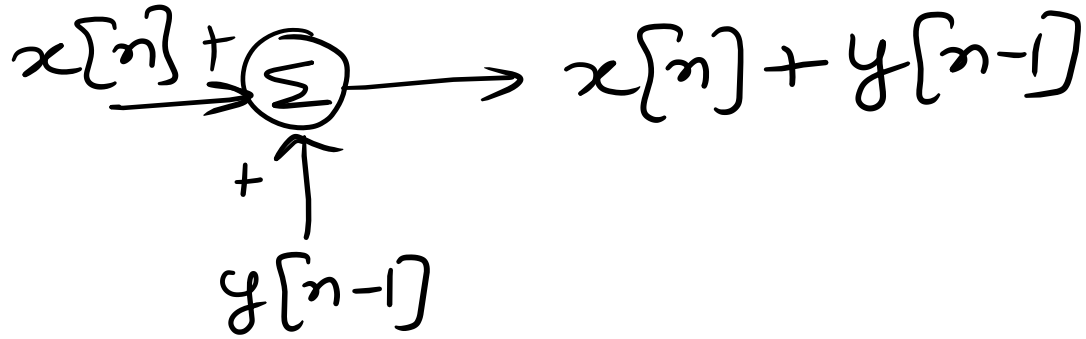
$h[n] = \frac{1}{3}\delta[n] + \frac{1}{2}\delta[n-1] + 5\delta[n-2]$

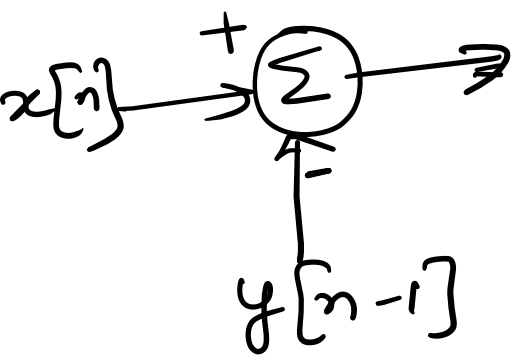


Since the duration of $h[n]$ is finite, it is called as the finite impulse response system or an FIR system.. This system is always stable.

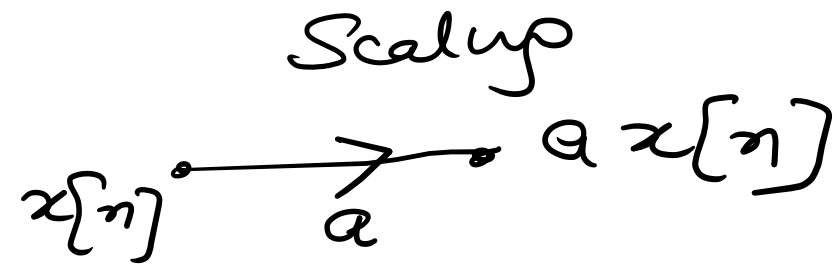
Block diagram representation

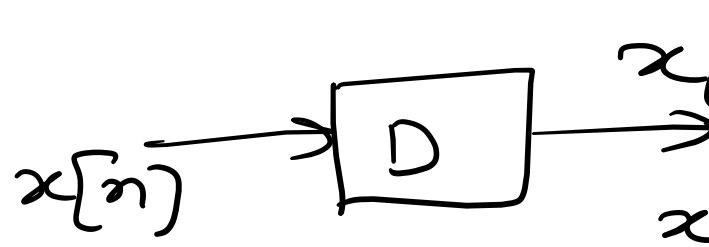
Adder





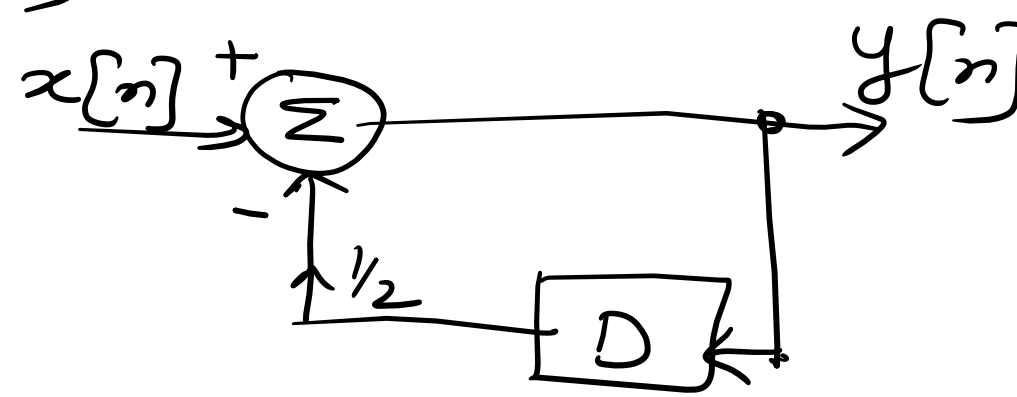
Scaling





$$y[n] + \frac{1}{2} y[n-1] = x[n]$$

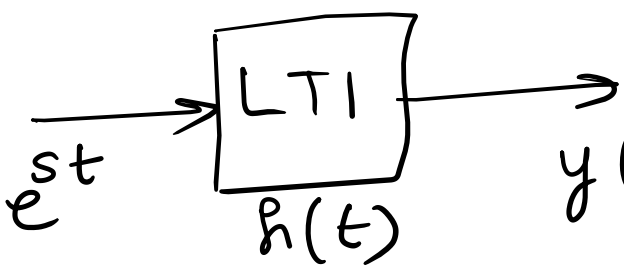
$$y[n] = x[n] - \frac{1}{2} y[n-1]$$



Fourier Series.

Analogy
A ≡ LTI system
v ≡ $x(t) = e^{st}$
λ ≡ H(s)

CT



S is some constant

$s \in \mathbb{C}$

$Av = \lambda v$

↑ eigenvector, eigenvalue

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau)\, e^{s(t-\tau)}\, d\tau$$

$$y(t) = e^{st} \int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

↓ H(s)

$$H(s) \triangleq \int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

then

$$y(t) = e^{st}\, H(s)$$

↑ fn. of t

↑ e^{st} acts as eigenfunctions of an LTI system, where H(s) is the corresponding eigenvalue.

s = purely imaginary

Let $s = j\omega$

$x(t) = e^{st}$

$= e^{j\omega_0 t}$

→ [LTI] →

$y(t) = H(s)\, e^{st}$

$= H(j\omega_0)\, e^{j\omega_0 t}$

o/p

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0)\, e^{jk\omega_0 t}$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

i/p to an LTI system

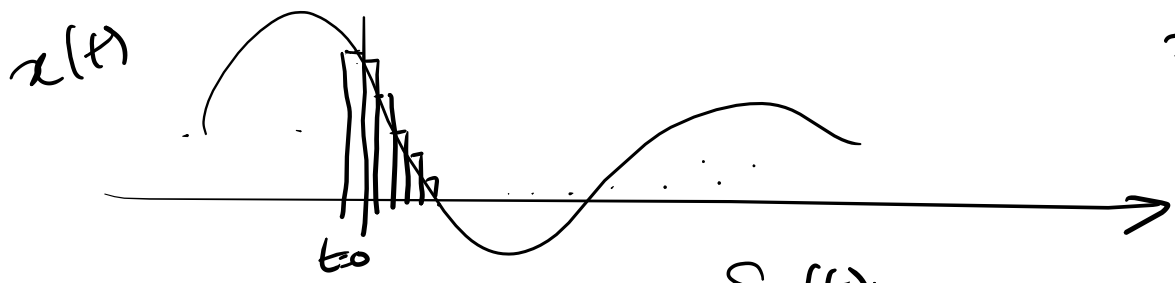
where $x(t)$ is periodic signal

ω_0 corresponds to the fundamental freq.

or $\omega_0 = \frac{2\pi}{T}$

$T \equiv$ fundamental time period.

F.S. representation of a C.T. signal.



$$x(t) \delta_\Delta(t) \Delta = x(0) \delta_\Delta(t) \Delta$$

$$\vdots$$

$$\vdots \quad x(t) \delta_\Delta(t-\Delta)\Delta = x(\Delta)\delta_\Delta(t)\Delta$$

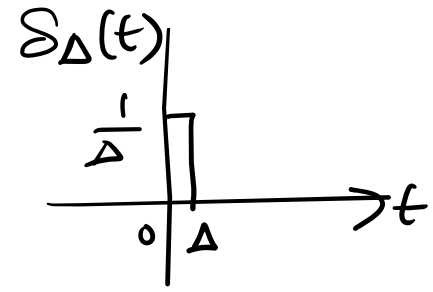
$$\vdots$$

$$x(t)\delta_\Delta(t-k\Delta)\Delta = x(k\Delta)\delta_\Delta(t)\Delta$$

$$\vdots$$

$$x(t) * \delta(t) = x(t)$$

$$= \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$$



$$\lim_{\Delta\to 0} \sum_{k=-\infty}^{\infty} x(k\Delta)\,\delta_\Delta(t-k\Delta)\,\Delta \longrightarrow \lim_{\Delta\to 0} \sum_{k=-\infty}^{\infty} x(k\Delta)\,h_\Delta(t-k\Delta)\Delta$$

$$= \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau \xrightarrow{\text{LTI}} \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

C.T. Signals.

F.S representation of a periodic signal is given by

Synthesis eqns.

$$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad ; \text{ where } \omega_0 = \frac{2\pi}{T}$$

$T \equiv$ fundamental time period of signal $x(t)$.

Given an LTI system

$$\Rightarrow y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

how the exponentials are modified in amplitude....

D.T. case

$x[n] = z^n$ → [LTI system] → $y[n] = ?$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, z^{n-k}$$

$$= z^n \boxed{\sum_{k=-\infty}^{\infty} h[k] z^{-k}}$$

$$= z^n H(z)$$

$$= H(z)\, z^n$$

Signals of the form z^n act as eigenfunctions of a D.T. LTI system & the corresponding eigenvalues are denoted by

$$H(z) \triangleq \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

if $z = e^{j\frac{2\pi}{N}}$

then if $x[n] = z^n = \left(e^{j\frac{2\pi}{N}}\right)^n = e^{j\frac{2\pi n}{N}}$

then o/p $y[n] = H\left(e^{j\frac{2\pi}{N}}\right) e^{j\frac{2\pi}{N}n}$

Let i/p $x[n] = \sum_{k=0}^{NH} a_k e^{jk\frac{2\pi}{N}n}$

then o/p $= \sum_{k=0}^{N-1} a_k H\left(e^{j\frac{k2\pi}{N}}\right) e^{jk\frac{2\pi}{N}n}$

∴ in discrete time, unique frequencies exist for any periodic signal between 0 to 2π interval.